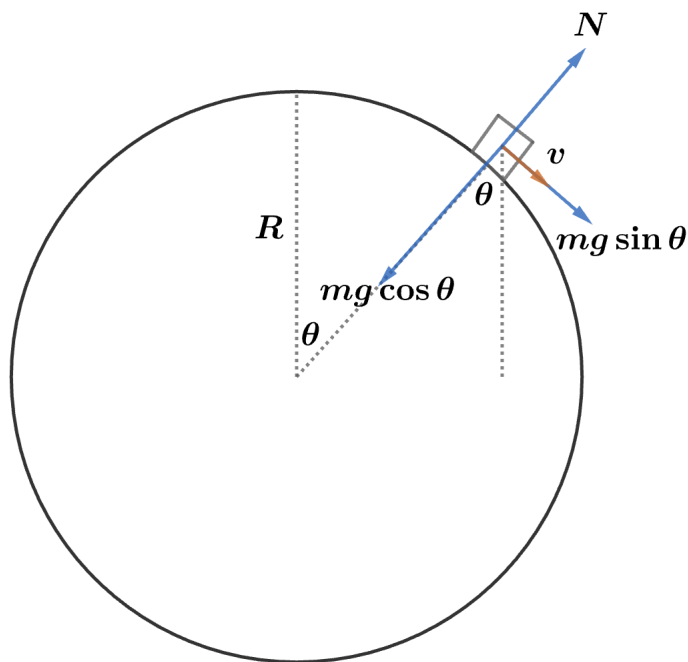


2016 F=ma Exam: Problem 9

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If the bead leaves the sphere at angle θ , we can find its speed by conservation of energy. Setting the potential energy reference at the center of the sphere, we have

$$E_i = mgR$$
$$E_f = \frac{1}{2}mv^2 + mgR \cos \theta$$

Since $E_i = E_f$,

$$mgR = \frac{1}{2}mv^2 + mgR \cos \theta$$
$$v^2 = 2gR(1 - \cos \theta)$$

To find θ , we note the bead is undergoing circular motion and apply Newton's 2nd law,

$$mg \cos \theta - N = \frac{mv^2}{R}$$

At the instant the bead leaves the sphere, $N \rightarrow 0$ so

$$mg \cos \theta = \frac{mv^2}{R}$$

$$\cos \theta = \frac{v^2}{gR}$$

Plugging this into our earlier equation,

$$v^2 = 2gR(1 - v^2/gR) = 2gR - 2v^2$$

$$3v^2 = 2gR$$

$$v = \sqrt{\frac{2gR}{3}} = \sqrt{\frac{gD}{3}}$$

using the fact that $D = 2R$. Thus, the answer is E.